Acidity, Basicity, and p*K*a

- **Why some molecules are acidic and others basic**
- **Why some acids are strong and others weak**
- **Why some bases are strong and others weak**
- **Estimating acidity and basicity using pH and p*K*a**
- **Structure and equilibria in proton transfer reactions**
- **Which protons in more complex molecules are more acidic**
- **Which lone pairs in more complex molecules are more basic**

Acids and Bases: Definitions

- **Acids give protons**
- **Bases accept protons**



Brønsted –Lowry definition

Acids and Bases: Definitions

- **Acids give protons**
- **Bases accept protons**



Brønsted –Lowry definition

Bronsted acids are proton donors

$HA \longrightarrow A^- + H^+$

The dissociation yields a proton and the conjugate base (an anion, if the acid is a neutral compound) of the acid

An acid is a species having tendency to lose/give a proton

Acids and Bases: Definitions

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- **Bases accept protons**

→ **Brønsted –Lowry definition**

Bronsted acids are proton donors

HA ⟶ A^- + H^+

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An acid is a species having tendency to lose/give a proton

Bronsted bases are proton acceptors

B: + H^+ ⟶ B-H^+

The only way a Bronsted base can "accept" a proton is to supply an electron pair and form a bond to the proton.

A base is a species having tendency to accept a proton

Acids and Bases: Definitions

- **Acids give protons**
- **Bases accept protons**



Brønsted –Lowry definition

The two equations are not strictly correct because a Bronsted acid does not dissociate, it donates a proton to something which accepts a proton.

The proton does not just dissociate and float around in solution, it is always attached to something

Acids and Bases: Definitions

- **Acids accept electrons**
- **Bases give electrons**



Lewis definition

A Lewis acid is an electron acceptor
A Lewis base is an electron donor

Acids and Bases: Definitions

- **Acids accept electrons**
- **Bases give electrons**



Lewis definition

A Lewis acid is an electron acceptor
A Lewis base is an electron donor

This is a more general definition as this definition includes the proton (as it can accept electron pairs).
Common Lewis acids: BF_3, $TiCl_4$, $SnCl_4$, $AlCl_3$

All accept unshared or shared electron pairs (not tightly bound such as π electrons)

Acid-base reactions

- **No compound is inherently an acid or a base.**

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- **A compound only functions as an acid if it donates a proton to something**

Acid-base reactions

- **No compound is inherently an acid or a base.**
- **A compound only functions as an acid if it donates a proton to something**
- **There must be a base to have an acid-base reaction**
  - **HCl**  **$H^+ + Cl^-$ Vapor phase- no reaction**

    $\Delta G°_{298K} = +1347\ kJ.mol^{-1}$

But in water **$HCl + H_2O$**  **$H_3O^+ + Cl^-$**

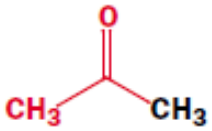
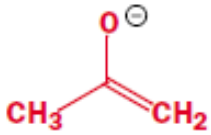
Acids and Bases: Examples

HCl H_2SO_4

CH_3COOH

NH_4^+

$AlCl_3$ $FeCl_3$

Acid	pK_a	Conjugate base
HI	ca. –10	I^-
HCl	ca. –7	Cl^-
H_2SO_4	ca. –3	HSO_4^-
HSO_4^-	2.0	SO_4^{2-}
CH_3COOH	4.8	CH_3COO^-
H_2S	7.0	HS^-
NH_4^+	9.2	NH_3
C_6H_5OH	10.0	$C_6H_5O^-$
CH_3OH	15.5	CH_3O^-
	20.0	
	25	
NH_3	33	NH_2^-
C_6H_6	ca. 43	$C_6H_5^-$
CH_4	ca. 48	CH_3^-

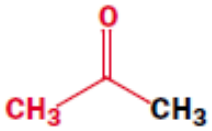
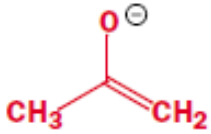
Acids and Bases: Examples

HCl H_2SO_4 NaOH KOH

CH_3COOH NH_3 NH_2^-

NH_4^+ CH_3COO^-

$AlCl_3$ $FeCl_3$

Acid	pK_a	Conjugate base
HI	ca. –10	I^-
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H_2SO_4	ca. –3	HSO_4^-
HSO_4^-	2.0	SO_4^{2-}
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H_2S	7.0	HS^-
NH_4^+	9.2	NH_3
C_6H_5OH	10.0	$C_6H_5O^-$
CH_3OH	15.5	CH_3O^-
	20.0	
	25	
NH_3	33	NH_2^-
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CH_4	ca. 48	CH_3^-

Acid Strength

- **How to quantify the strength of an acid?**
  - **H_2SO_4 (battery acid) is stronger than CH_3CO_2H (vinegar)**
  - **Sulfuric acid is a better proton donor than acetic acid**

Acid Strength

- **How to quantify the strength of an acid?**
  - **H_2SO_4 (battery acid) is stronger than CH_3CO_2H (vinegar)**
  - **Sulfuric acid is a better proton donor than acetic acid**

  - **To compare acid let's compare their equilibrium in water**

    - $HA + H_2O \overset{K_{eq}}{\rightleftharpoons} H_3O^+ + A^-$ K_{eq}

Acid Strength

- $K_{eq.} = \dfrac{[H_3O^+][A^-]}{[HA][H_2O]}$

- $K_{eq.}[H_2O] = K_a = \dfrac{[H_3O^+][A^-]}{[HA]}$

- $pK_a = -\log K_a$

Acid Strength

- **Acetic acid pK_a = 4.75 = -log K_a (K_a = 1.8 10^{-5}) *vs.***
  Hydrofluoric acid pK_a = 3.2 = -log K_a(K_a = 6.3 10^{-4})

HF is stronger than CH_3CO_2H by 1.5 order of magnitude

Acid Strength

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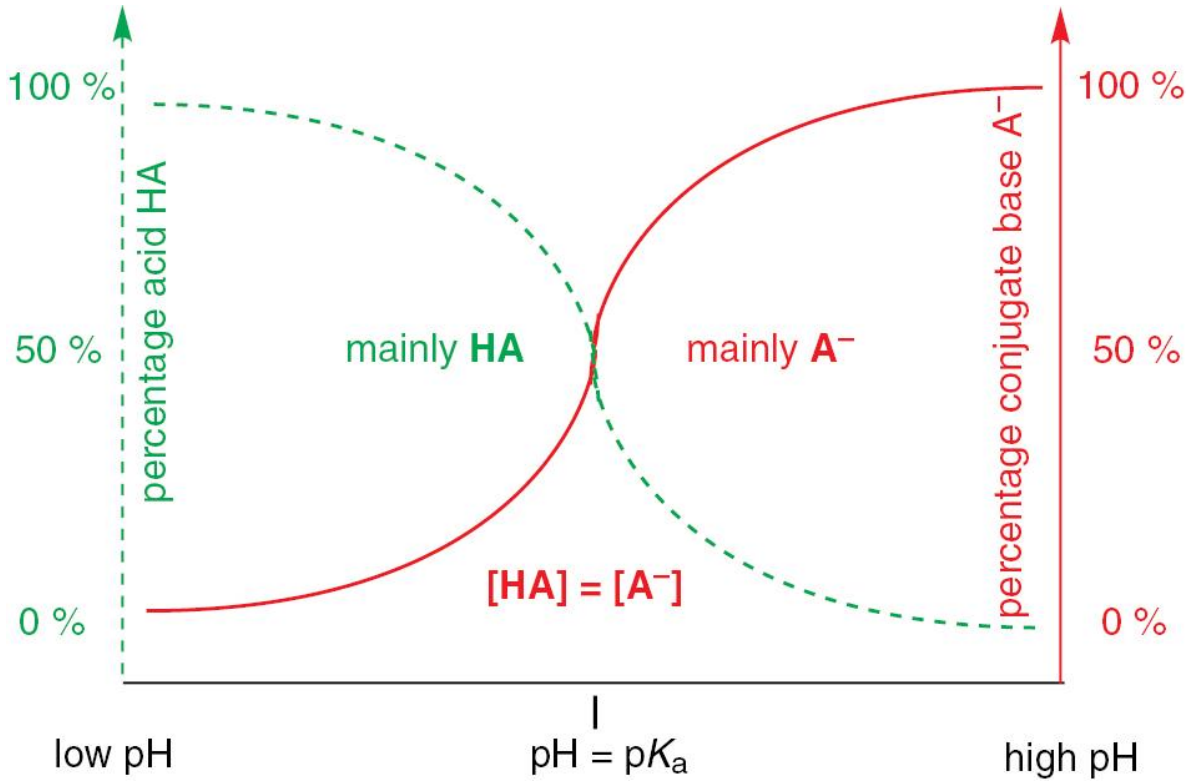
  HF is stronger than CH_3CO_2H by 1.5 order of magnitude

- **Acetic acid pK_a = 4.75 = -log K_a (K_a = 1.8 10^{-5}) *vs.***
  Hydrochloric acid pK_a = -6.6 = -log K_a (K_a = 3.98 10^{6})
  HCl is stronger than CH_3CO_2H by 10 order of magnitude

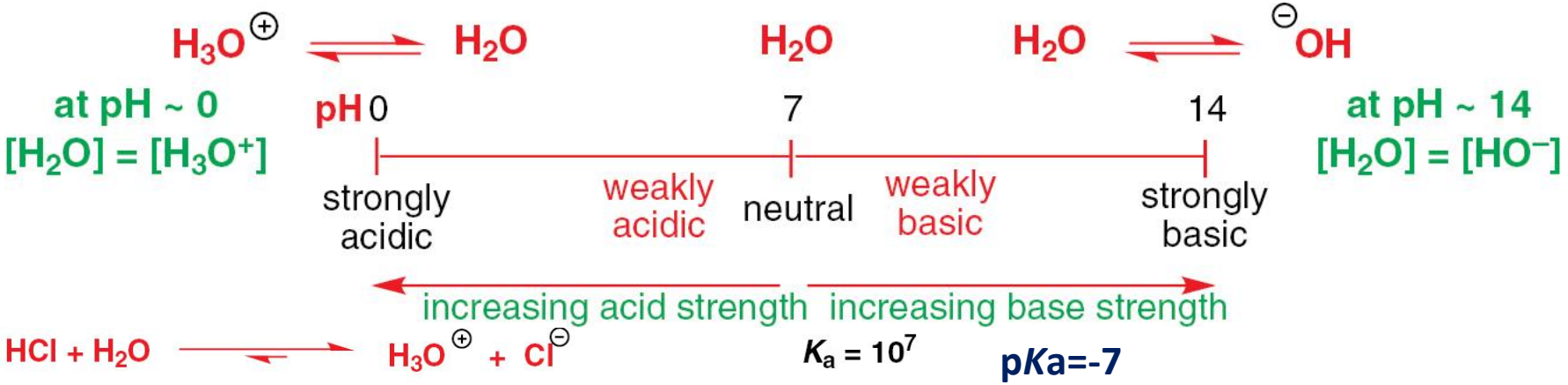
Definition of p*K*a-How is it related to pH?

- **The larger the *K*a, the lower the p*K*a (minus sign): the stronger the acid**

- **The p*K*a of the acid is the pH at which it is half-dissociated**



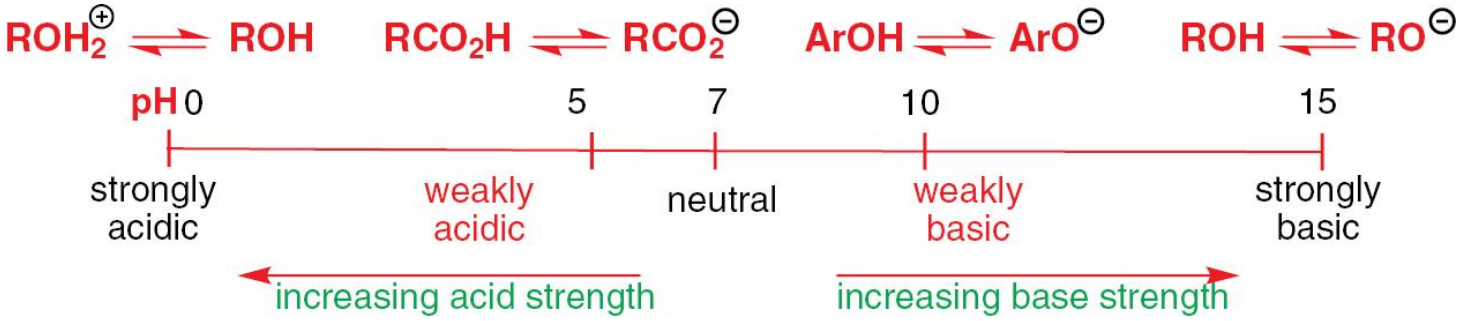
Examples on p*K*a-pH



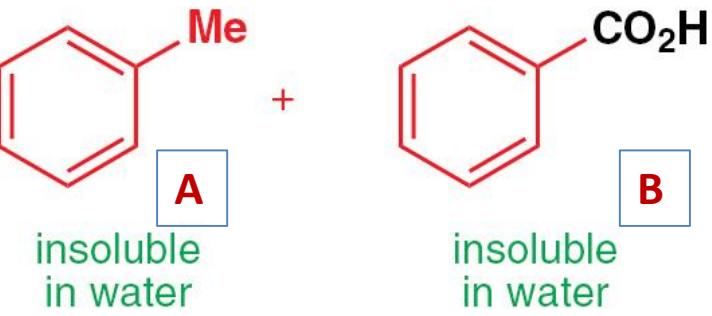
$HCl + H_2O \rightleftharpoons H_3O^{\oplus} + Cl^{\ominus}$ $K_a = 10^7$ **p*K*a=-7**



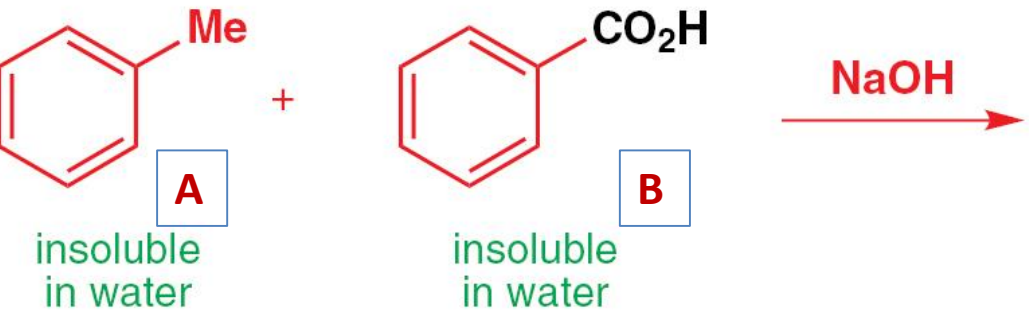
$K_a = 1.74 \times 10^{-5}$ **p*K*a=4,8**



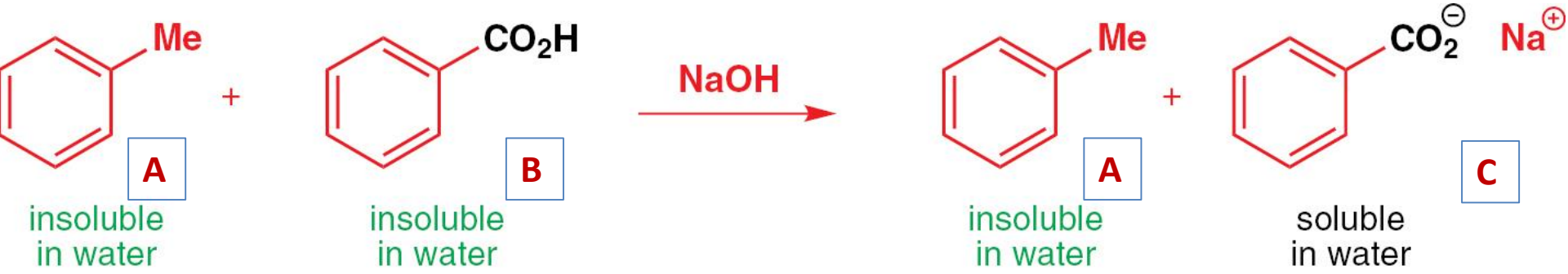
Practical Applications: Extraction



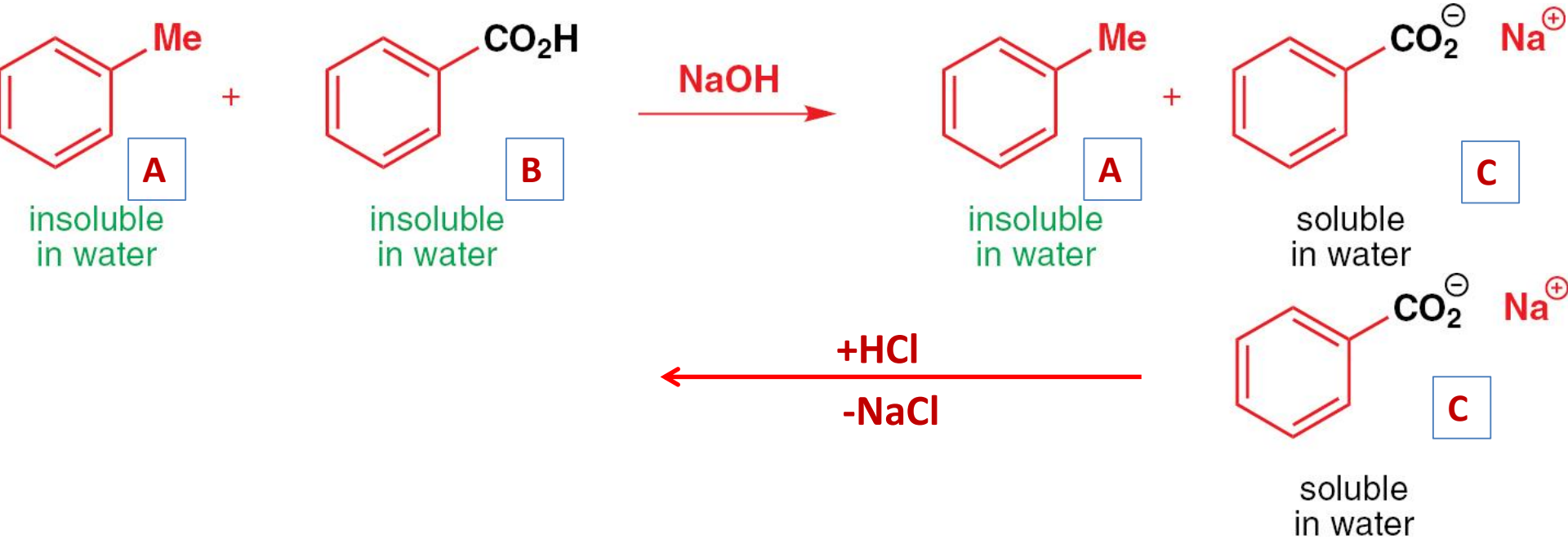
Practical Applications: Extraction

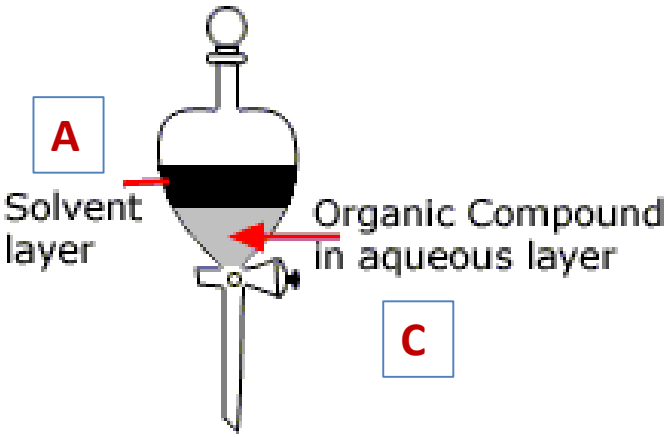


Practical Applications: Extraction

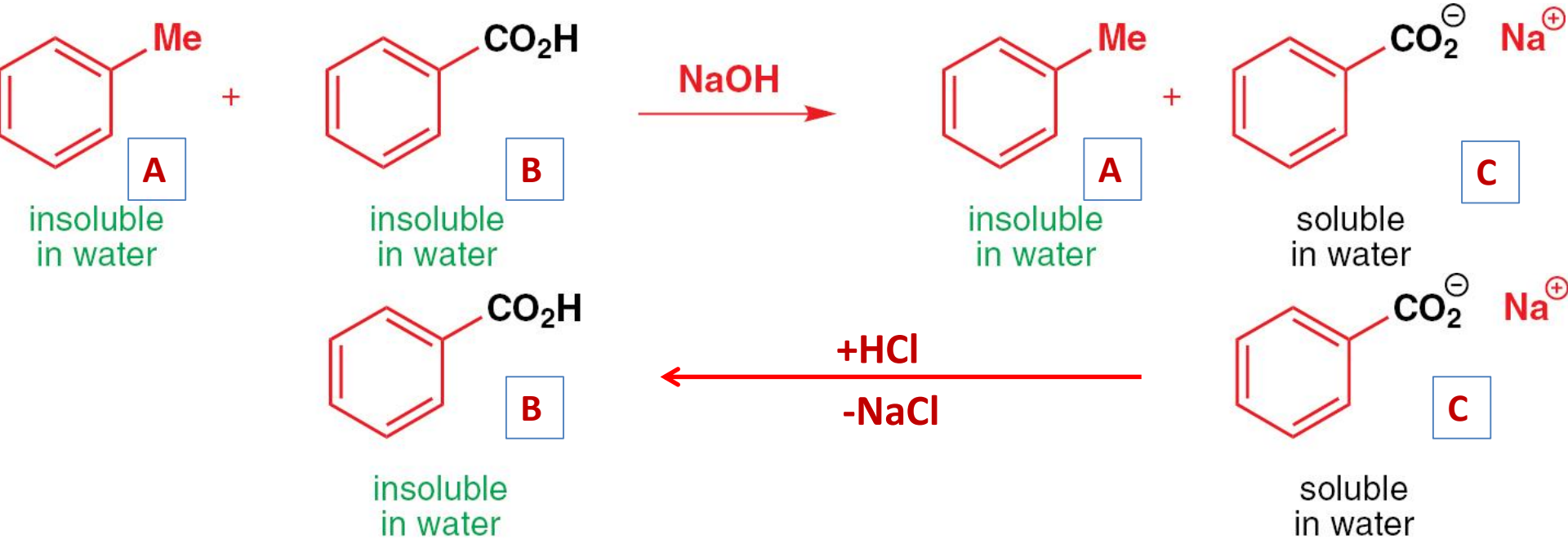


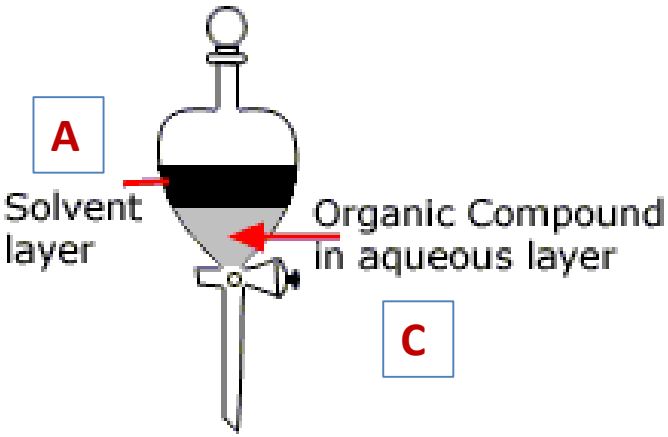
Practical Applications: Extraction





Practical Applications: Extraction





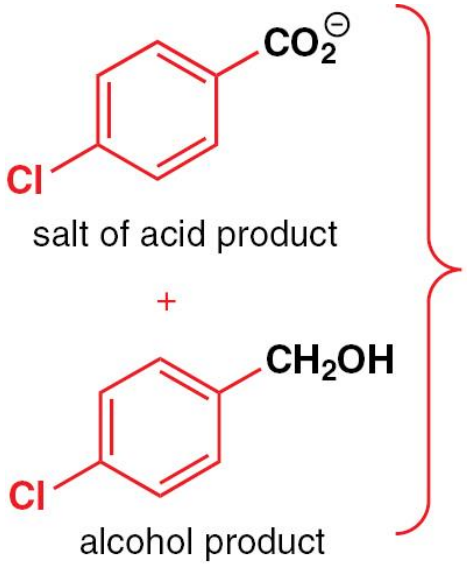
Practical Applications: Extraction



starting material

acid product

alcohol product

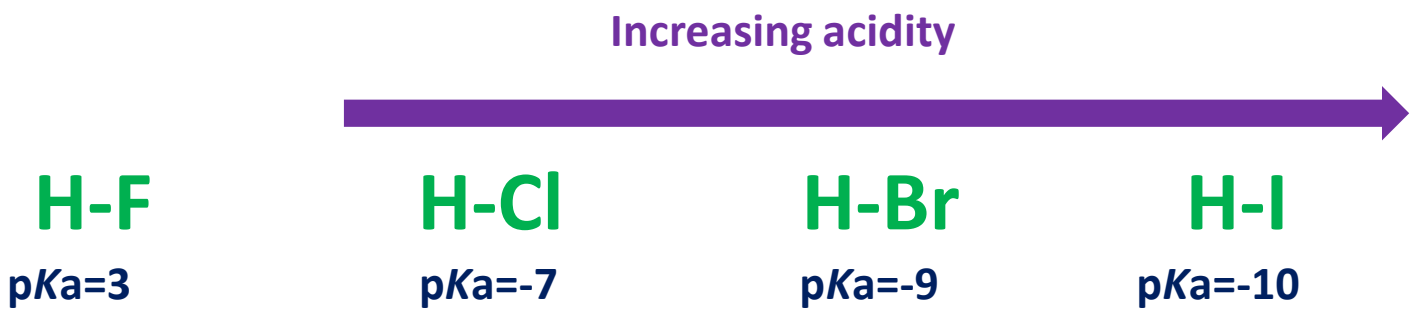


Factors affecting acidity

1. The strength of the HA bond
2. The ability of the conjugate base to accommodate the negative charge either by delocalization or inductive effects
3. Structural features-functional groups
4. Hybridization state

Strength of the HA bond

The weaker the bond strength the greater the acidity:
H^+ more easily removed



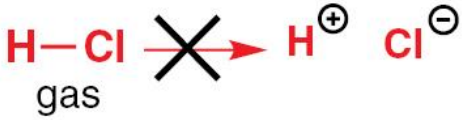
Strength of the HA bond

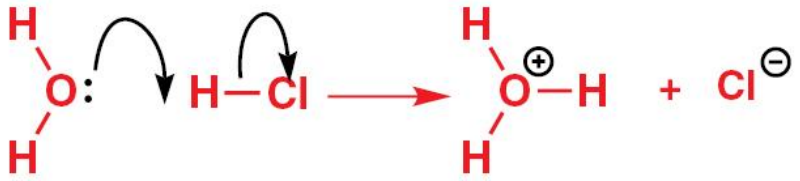
The weaker the bond strength the greater the acidity:
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Increasing acidity

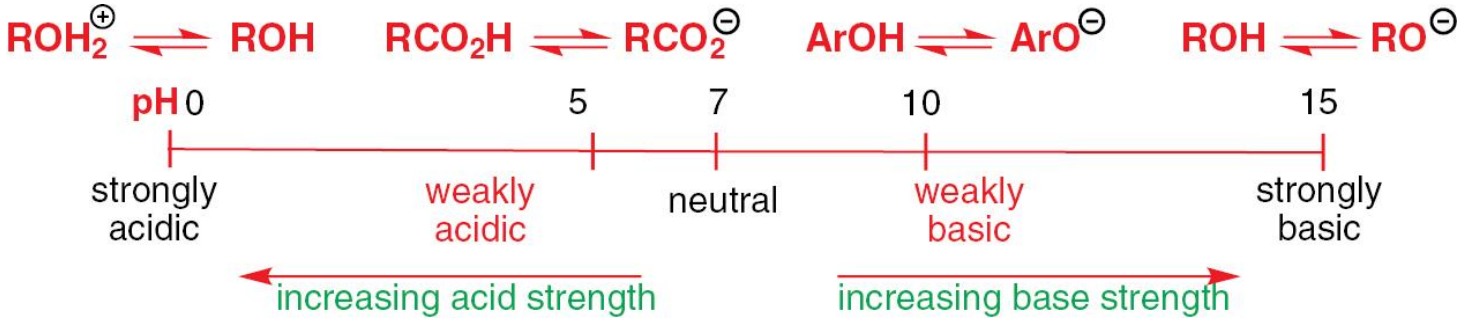
H-F	H-Cl	H-Br	H-I
p*K*a=3	p*K*a=-7	p*K*a=-9	p*K*a=-10

Increasing bond strength



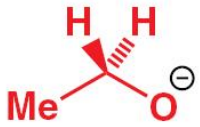


Stabilization of the conjugate base by delocalization



The greater the delocalization of the negative charge of the conjugate base, the stronger the acid

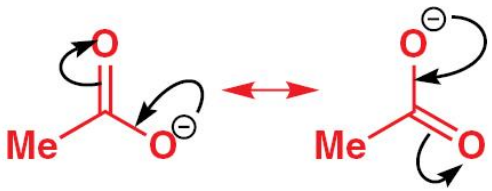
ethoxide



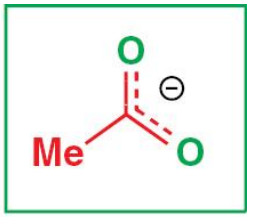
charge localized on one oxygen

p*K*a=15.9

acetate



charge delocalized over two oxygens



p*K*a=4.8

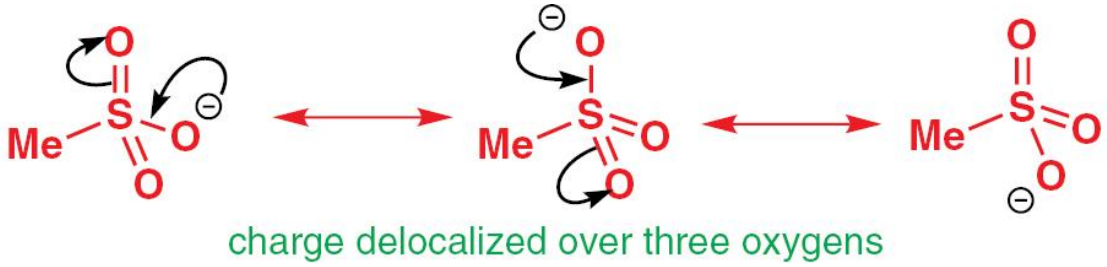
Localized charge

Delocalization over 2 O-atoms

Stabilization of the conjugate base by delocalization

The greater the delocalization of the negative charge of the conjugate base, the stronger the acid

methanesulfonate

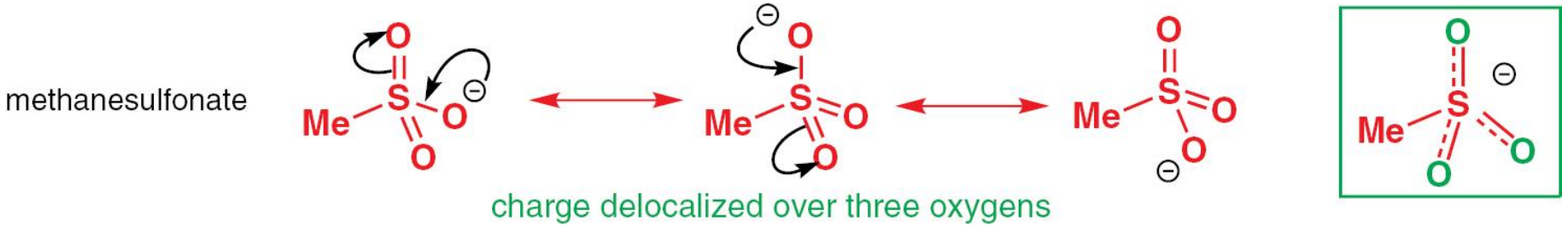


charge delocalized over three oxygens

p*K*a=-1.9

Stabilization of the conjugate base by delocalization

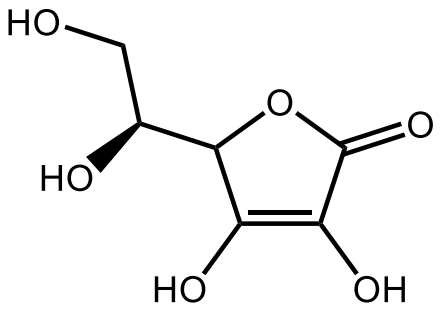
The greater the delocalization of the negative charge of the conjugate base, the stronger the acid



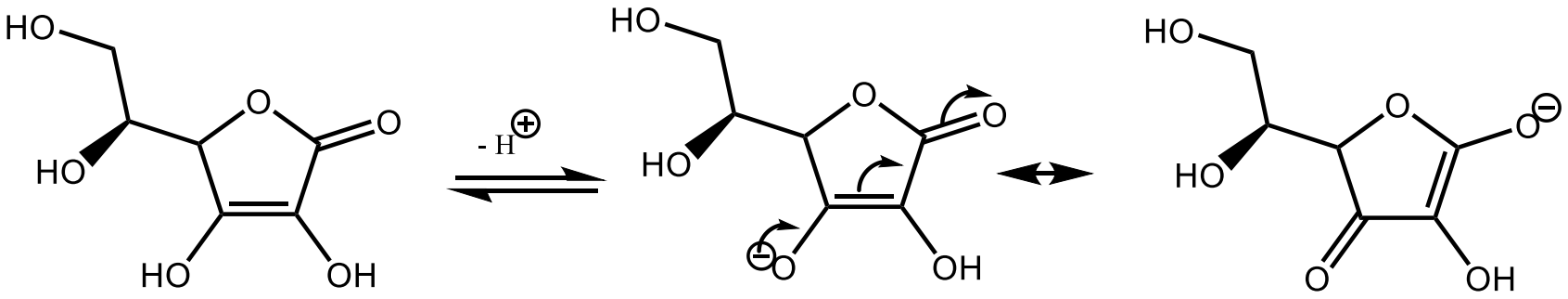
p*K*a=-1.9

Delocalization over 3 O-atoms

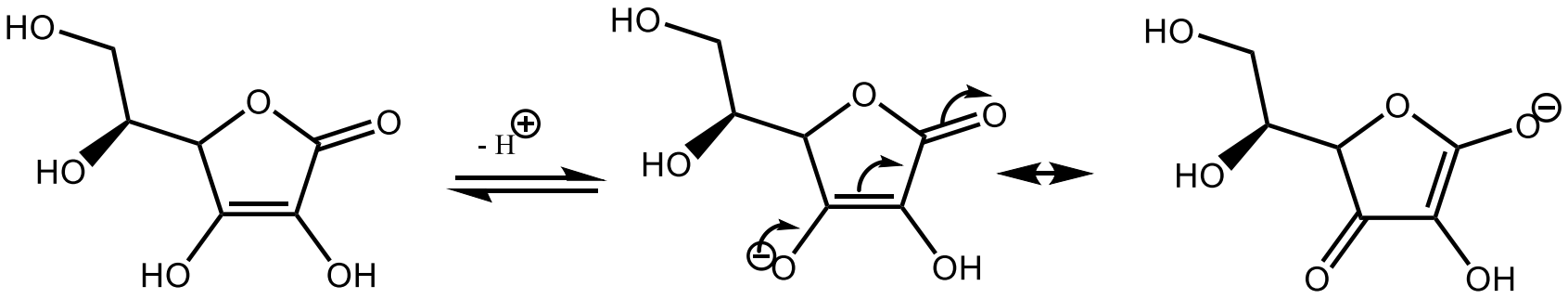
Is Vitamin C acidic?



Is Vitamin C acidic?

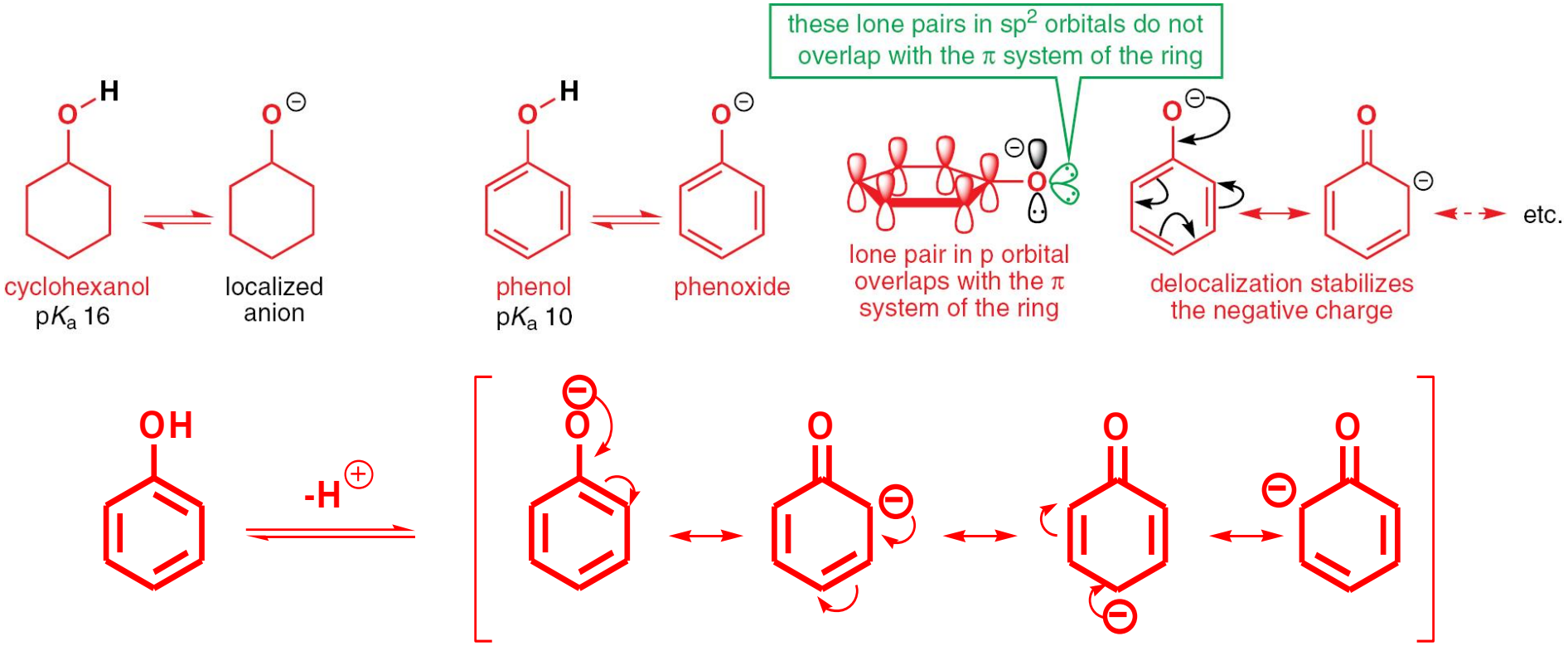


Is Vitamin C acidic?



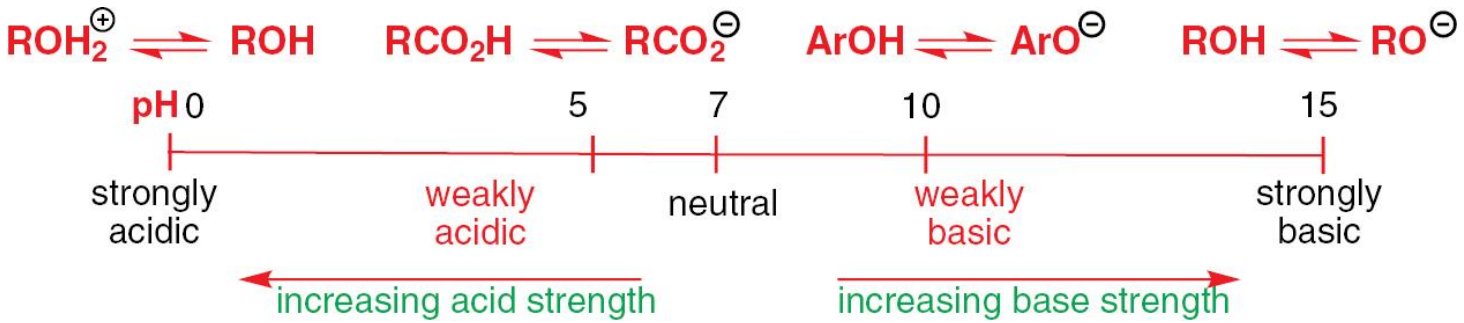
p*K*a=4.2, lower than p*K*a of AcOH!

The acidity of Phenols



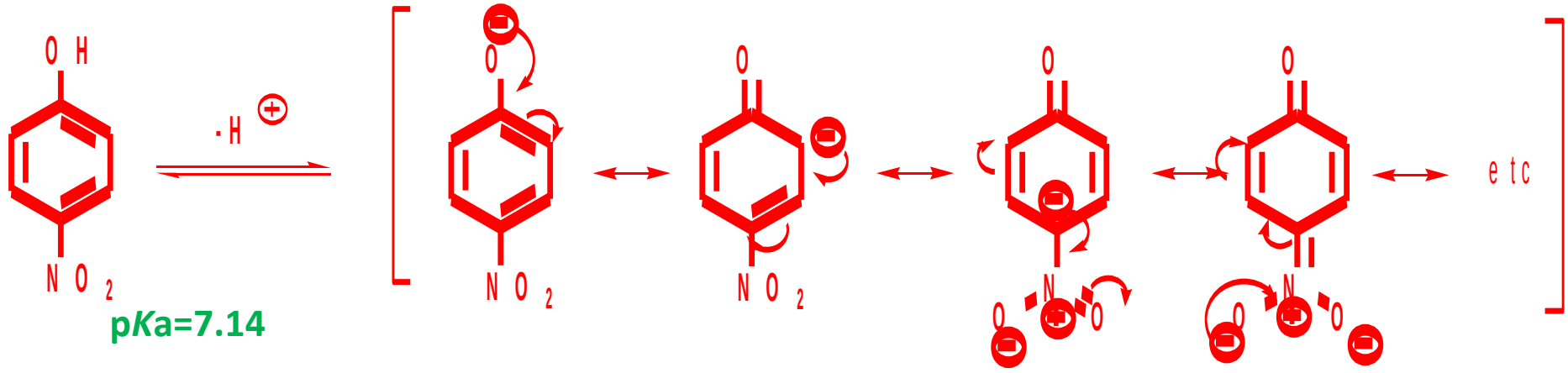
Delocalization of the negative charge of the phenoxide anion:
Notice that the charge is delocalized in *ortho* and *para* positions?

The acidity of some organic compounds



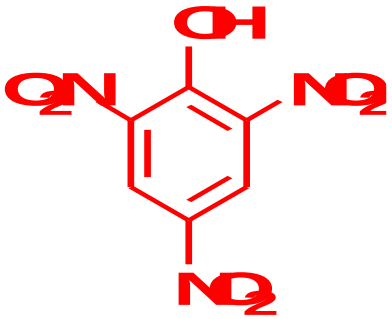
The weaker the acid, the stronger the base required to remove the proton

The acidity of Nitrophenols



p*K*a=7.14

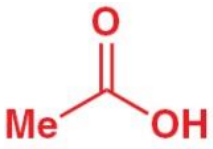
The NO_2 group is strongly electron-withdrawing and can stabilize the anion by both delocalization and inductive effects.



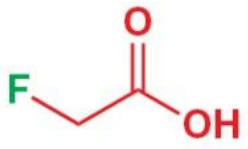
2,4,6-trinitrophenol is strongly acidic (and explosive): p*K*a=0.7!

Stabilization of the conjugate base by inductive effects

The Inductive effect (I) is the transmission of charge through sigma bonds



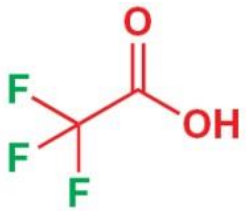
acetic acid



fluoroacetic acid



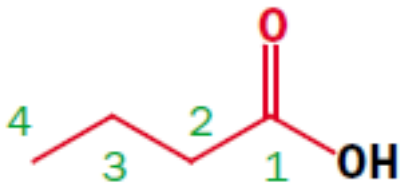
difluoroacetic acid



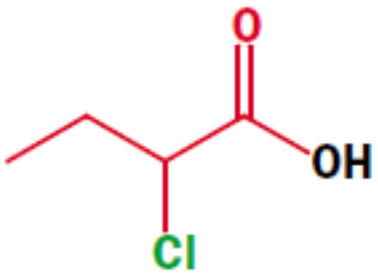
trifluoroacetic acid

Electronegative atoms can stabilize the charge of the conjugate base by withdrawing electron density

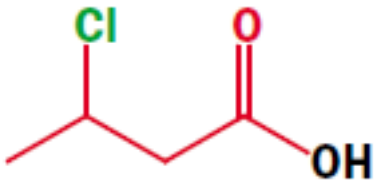
Inductive effects decrease when groups move apart



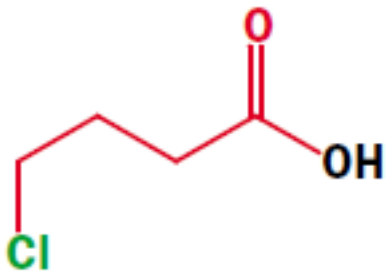
pK_a 4.8



pK_a 2.8



pK_a 4.1

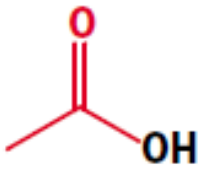


pK_a 4.5

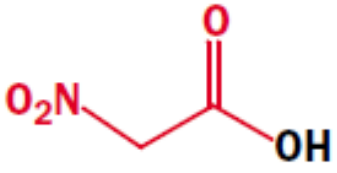
Notice that 3-chloro butanoic acid and 4-chloro butanoic acid have similar p*K*a to butanoic acid?

Electron-withdrawing groups: Examples

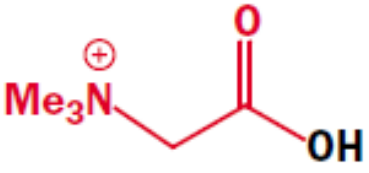
electron-withdrawing groups lowering the pK_a of carboxylic acids



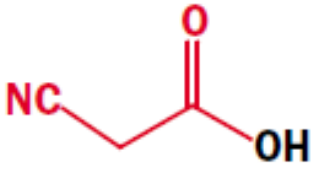
pK_a 4.76



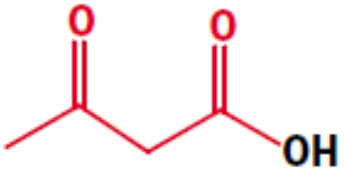
pK_a 1.7



pK_a 1.8



pK_a 2.4



pK_a 3.6

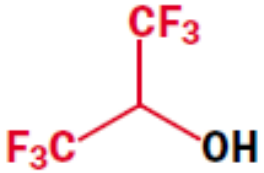
electron-withdrawing groups lowering the pK_a of alcohols



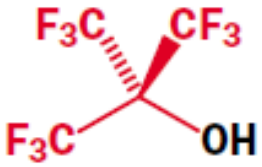
pK_a 15.5



pK_a 12.4



pK_a 9.3

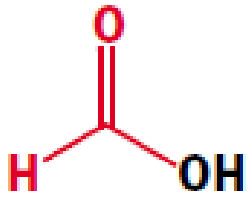


pK_a 5.4

Electron-donating groups decrease acidity

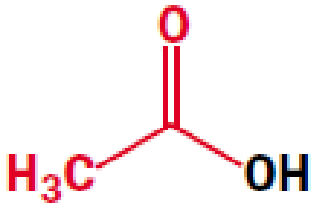
Alkyl groups are electron-donating groups: They decrease acidity

EXAMPLES



formic (methanoic) acid

pK_a 3.7



acetic (ethanoic) acid

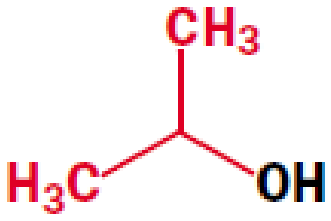
pK_a 4.8



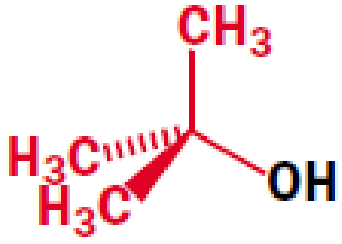
pK_a 15.5



pK_a 16.0

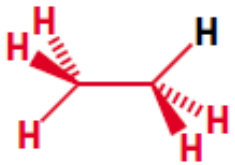


pK_a 17.1



pK_a 19.2

Hybridization state affects acidity



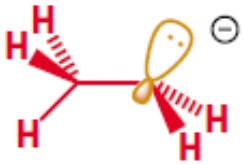
pK_a ca. 50



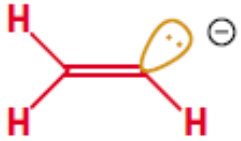
pK_a ca. 44



pK_a ca. 26



lone pair of $CH_3CH_2^{\ominus}$ in sp^3 orbital

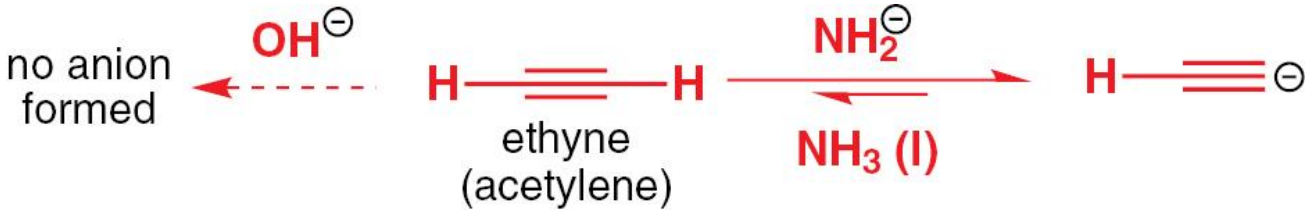


lone pair of $CH_2{=}CH^{\ominus}$ in sp^2 orbital



lone pair of $HC{\equiv}C^{\ominus}$ in sp orbital

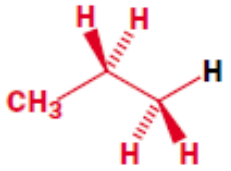
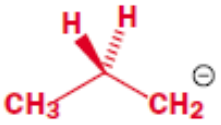
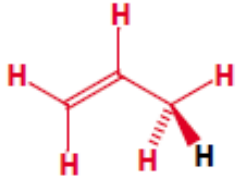
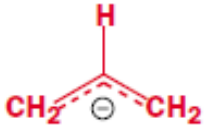
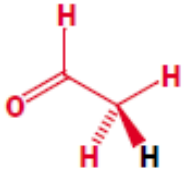
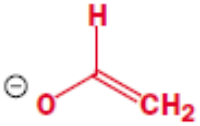
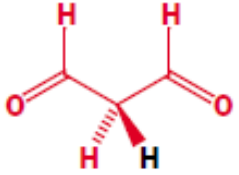
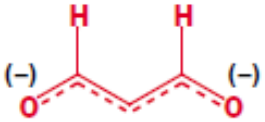
Acidity increases with increasing s character:
In an s orbital the electrons are closer to the nucleus



Carbon acids

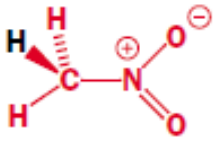
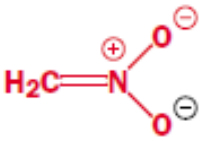
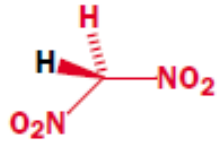
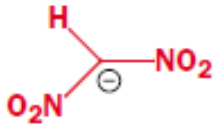
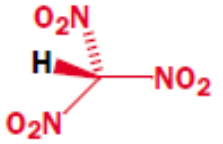
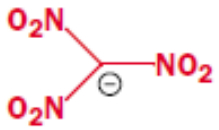
Carbon acids require a proton to be removed to form the carbanion (conjugate base): Since they are not very stable carbon acids are weak acids

EXAMPLES

Acid	Conjugate base	pK_a	Comments
		~50	charge is localized on one carbon—difficult since carbon is not very electronegative
		~43	charge is delocalized over π system—better but still not really good
		13.5	charge is delocalized over π system but is mainly on the electronegative oxygen—much better
		5	charge delocalized over π system but mainly over two oxygens—better still

Carbon acids-Stabilization by electron-withdrawing groups

EXAMPLES

Acid	Conjugate base	pK_a	Comments
		10	charge is delocalized but mainly on oxygens of nitro group
		4	charge can be delocalized over two nitro groups—more stable anion
		0	charge can be delocalized over three nitro groups—very stable anion

Electron-withdrawing groups stabilize the carbanion

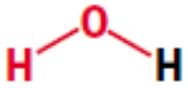
Stabilized carbanions are VERY important because they are often good nucleophiles for C-C bond formation

How to stabilize the conjugate base?

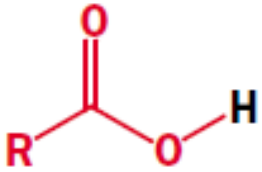
The more stabilized the conjugate base, A^-, the stronger is the acid, HA. Ways to stabilize A^- include:

- Having the charge on an electronegative element
- Delocalizing the negative charge over other carbon atoms, or even better, over more electronegative atoms
- Spreading out the charge over electron-withdrawing groups by the polarization of σ bonds (inductive)
- Having the negative charge in an orbital with more s character

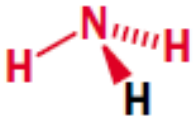
Acidity-Basicity



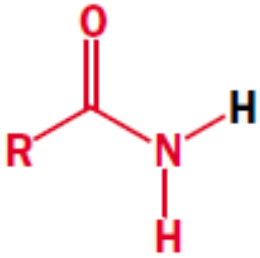
water
pK_a 15.74



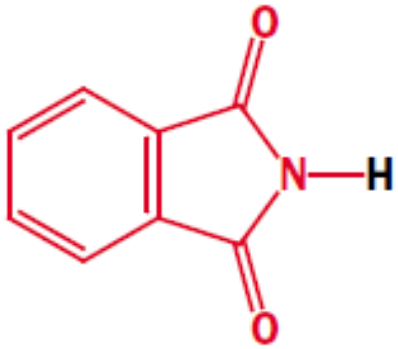
carboxylic acid
pK_a ca. 5



ammonia
pK_a ca. 33



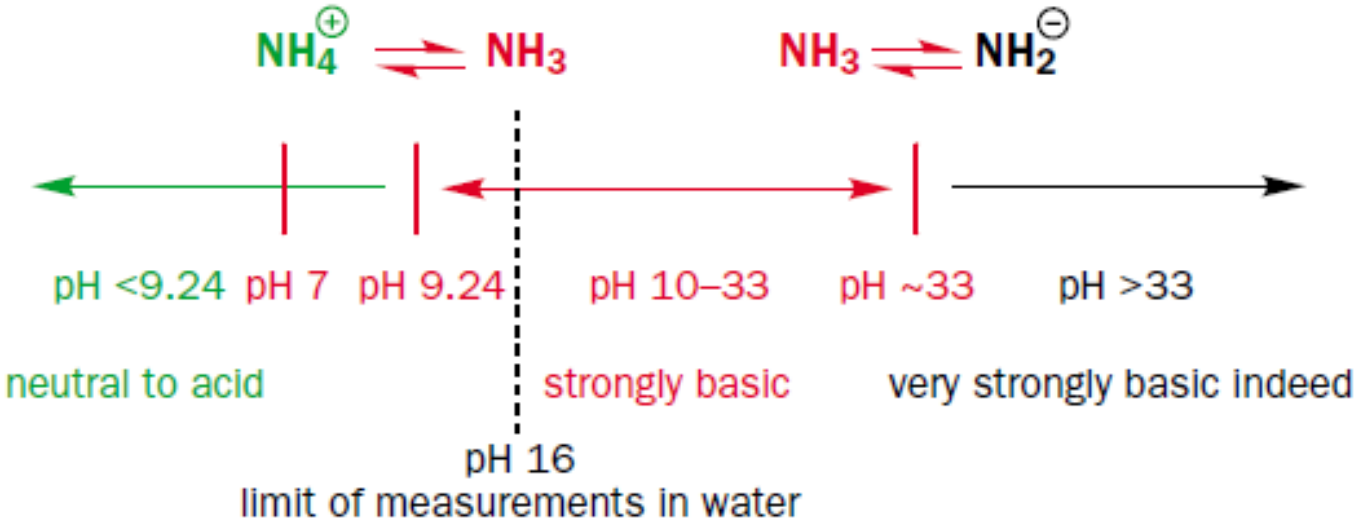
amide
pK_a ca. 17



phthalimide
pK_a 8.3

Amides are not so basic!

Basicity of NH_3



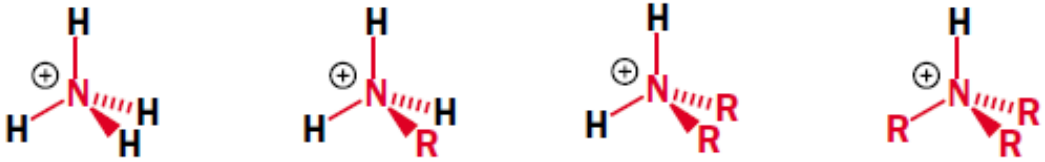
The higher the p*K*aH of the conjugate acid, the greater the basicity:

H_2O p*K*a=15.7
NH_4^+ p*K*a=9.24

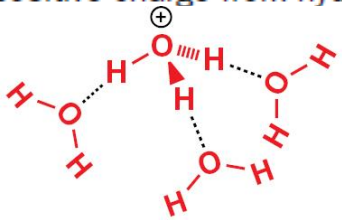
^-OH is a stronger base than NH_3

Basicity of amines

more stabilization of positive charge from alkyl groups



more stabilization of positive charge from hydrogen bonding with solvent



a structure for a solvated hydronium ion in water: the dashed bonds represent hydrogen bonds

pK_{aH} values for primary, secondary, and tertiary amines

R	pK_{aH} RNH_2	pK_{aH} R_2NH	pK_{aH} R_3N
Me	10.6	10.8	9.8
Et	10.7	11.0	10.8
n-Pr	10.7	11.0	10.3
n-Bu	10.7	11.3	9.9

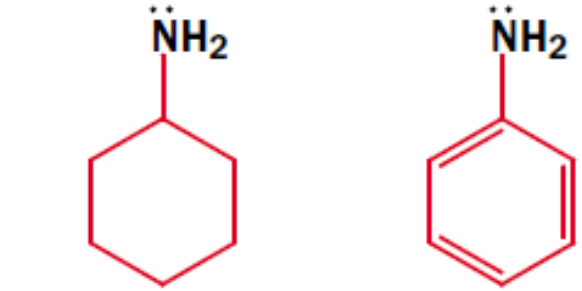
- All the amines have pK_{aH}s greater than that of ammonia (9.24)
- All the primary amines have approximately the same pK_{aH} (about 10.7)
- All the secondary amines have pK_{aH}s that are slightly higher
- Most of the tertiary amines have pK_{aH}s lower than those of the primary amines

Effects that decrease the electron density on the Nitrogen

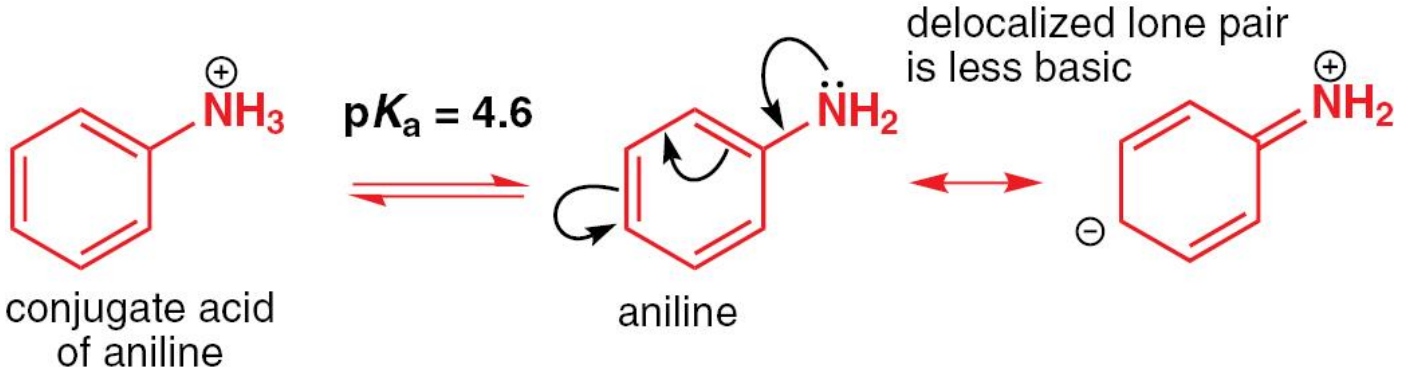
The lone pair on nitrogen will be *less* available for protonation, and the amine *less* basic, if:

- The nitrogen atom is attached to an electron-withdrawing group
- The lone pair is in an sp or sp^2 hybridized orbital
- The lone pair is conjugated with an electron-withdrawing group
- The lone pair is involved in maintaining the aromaticity of the molecule

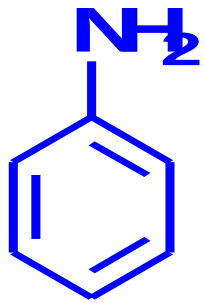
Effects that decrease the electron density on the Nitrogen

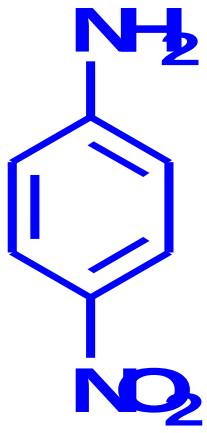


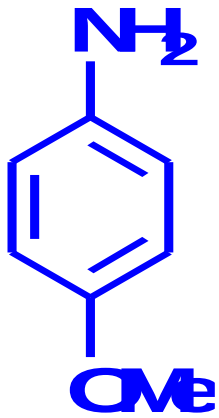
- The higher the p*K*aH of the conjugate acid, the greater the basicity
- Aniline is a much weaker base than cyclohexylamine
- The lone pair is delocalized in the benzene ring



Question: Which of these anilines is more basic?





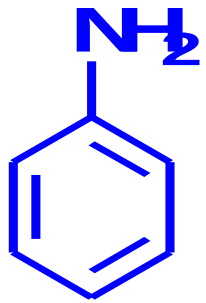


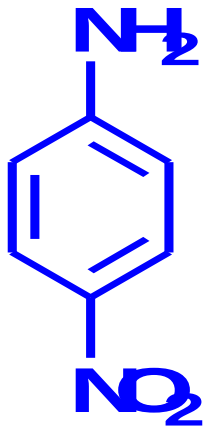
Electron-withdrawing effect

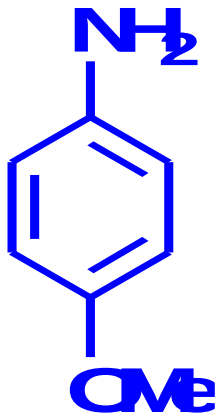
Electron-donating effect

Most basic

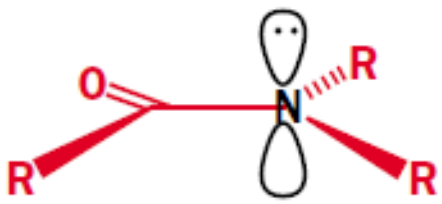
Question: Which of these anilines is more basic?



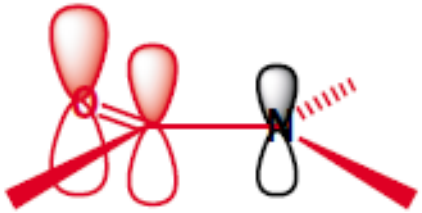




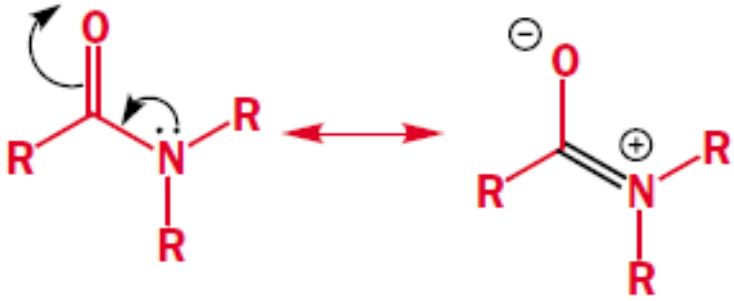
What about amides?



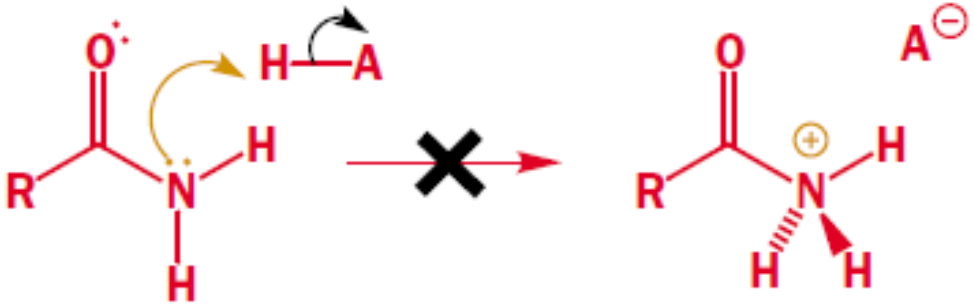
nitrogen is sp^2 hybridized with its lone pair in a p orbital



good overlap with the carbonyl group

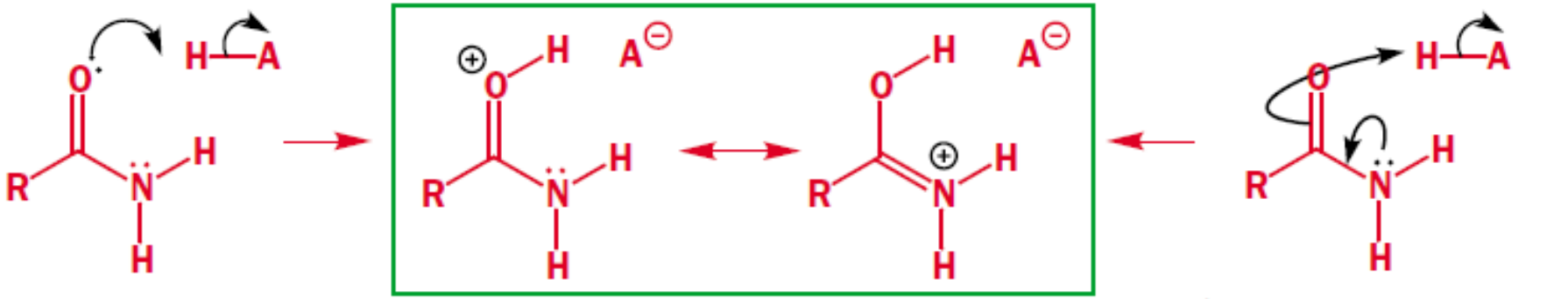


delocalization of nitrogen's lone pair into π system

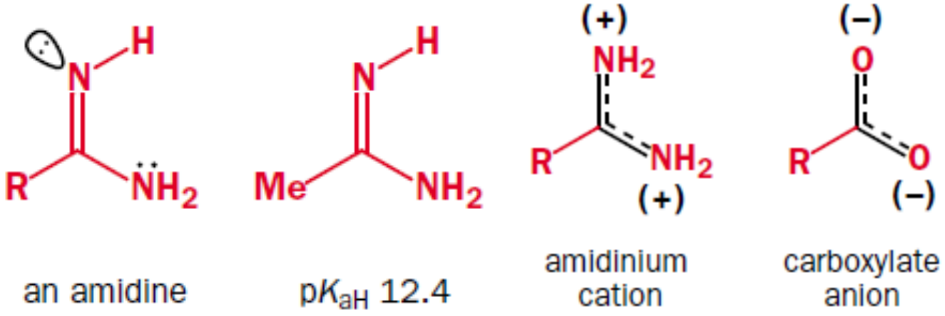


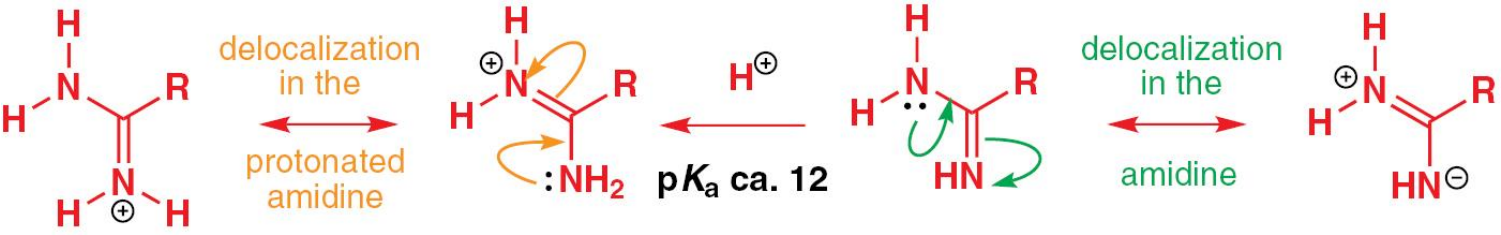
no protonation occurs on the nitrogen atom

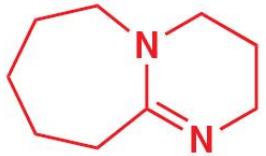
Amides


protonation occurs on the oxygen atom
these structures are just two different ways to draw the same delocalized cation
these arrows emphasise the contribution of the nitrogen's lone pair.

Amidines are more basic than amines







DBU
diazabicycloundecane

Conclusions

- Some molecules are acidic and others basic
- Some acids are strong and others are weak
- Estimating acidity and basicity using pH and p*K*a
- Factors affecting acidity
- Factors affecting basicity
- Which protons in more complex molecules are more acidic
- Which lone pairs in more complex molecules are more basic

- It is the stability of the conjugate base (or acid) that counts!!